

Outperformance of Mall-Receptionist Android as Inverse Reinforcement Learning is Transitioned to Reinforcement Learning

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Abstract—Robots can tackle human–robot interaction (HRI) tasks through inverse reinforcement learning (IRL). However, offline IRL agents’ performance is upper-bounded by experts. Limited demonstration fails to provide an overall picture of the environment, especially in real-world applications. To further enhance IRL’s performance, we implement a cross-modal inverse reinforcement learning to reinforcement learning (IRL-to-RL) transition framework for a real-world HRI interaction task, in which a mall receptionist android promotes sanitizer usage. During the 10-day experiment, the android develops a more proactive and effective strategy than the human expert. Furthermore, we explore four decay modes of prior knowledge supervision and suggest a preferable pattern for practical use. Our results demonstrate the feasibility of the framework to assist robots in switching to diverse modalities, learning incrementally with a sparse reward function, and eventually outperforming the human expert. We anticipate our framework to inspire more IRL-to-better learning paradigms, facilitating robots to outcompete teachers in more real-world HRI applications.

Index Terms—Social HRI, imitation learning, reinforcement learning.

I. INTRODUCTION

INVERSE reinforcement learning (IRL) is a feasible approach for robots to perform human–robot interaction (HRI) tasks. First, by learning from demonstration (LfD), IRL-driven robots can rapidly accomplish tasks with satisfying performance [1]. Second, IRL avoids the hassle of hand-crafting

an applicable reward function, unlike reinforcement learning (RL) [2]. In practice, recent studies reflect that IRL agents can handle HRI tasks in both simulated settings [3] and strictly controlled real-world scenarios [4]. To further investigate IRL’s potential, we conducted a field experiment and achieved near-expert performance in [5].

However, IRL-driven robots still struggle in real-world HRI tasks owing to the following reasons:

- Data collection: Unlike in simulated settings, data collection in real-world scenarios is expensive. Generally, the gathered small amount of data is insufficient to present a complete description of the environment.
- Sub-optimal demonstration: Maintaining optimal performance over the entire experiment is challenging, even for human specialists. Learning from such a sub-optimal demonstration may worsen robots’ performance.
- Upper-bounded performance: Offline IRL agents solely LfD without extra data sampling. This imposes strong constraints on robots, of which conservative behaviors tend to be generated. Hence, IRL’s performance is typically upper-bounded by the demonstrator [6].

Researchers will thus attempt to deploy IRL-driven robots to mitigate the adverse effects of the above-mentioned restrictions and ensure smooth interactions.

Robots aided by RL can surpass human specialists. Recently, some world-shattering cases have been reported, such as AlphaZero [7] and GT Sophy [8], in which RL-driven agents defeated human experts. In addition to remarkable accomplishments in simulations [9], the deployment of RL has been generalized to real-world scenarios [10] and the HRI field [11], [12]. Nevertheless, well-developed RL agents require deliberately hand-crafted reward signals and readily accessible resources for trial and error. Therefore, warm-starting RL with fast-to-learn methods such as IRL seems to be a viable option for developing fully functioning robots in real-world HRI tasks with limited access to data collection.

To leverage the benefits of IRL and RL, we proposed a cross-modal IRL-to-RL transition framework. We examined it in a real-world HRI task, in which an android robot acts as a mall receptionist to encourage hand sanitizer usage,¹ as depicted in Fig. 1. The framework consists of two stages:

- First, an intermediary model was gained by cross-modal knowledge distillation (KD) based on the IRL policy.
- Second, an RL policy inherited the intermediary model’s encoder and was trained with a sparse reward function.

¹This research was approved by the ethical review board of the Graduate School of Engineering Science, Osaka University. (R1-5-5).

Manuscript received 6 December 2022; accepted 10 April 2023. Date of publication 14 April 2023; date of current version 25 April 2023. This letter was recommended for publication by Associate Editor K. S. Haring and Editor G. Venture upon evaluation of the reviewers’ comments. This work was supported by JST, Moonshot R&D under Grant JPMJMS2011. (*Corresponding author: Zhichao Chen.*)

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the ethical review board of graduate school of engineering science, Osaka University under Application No. R1-5-5.

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This letter has supplementary downloadable material available at <https://doi.org/10.1109/LRA.2023.3267385>, provided by the authors.

Digital Object Identifier 10.1109/LRA.2023.3267385



Fig. 1. Android interacts with customers in the shopping mall.

Meanwhile, it also received decaying supervision from prior knowledge during the training process.

Fig. 2 illustrates the framework architecture. To assess it, we conducted a ten-day trial. In the first six days, we showed that the android learned incrementally and ultimately surpassed the human expert. For better use of the framework, four decay modes of prior knowledge supervision were investigated in the last four days. Our results indicate the framework's viability of further refining IRL-driven robots' skills in real-world HRI.

The rest of this letter is structured as follows. In Section II, the task settings and the previous results are described. In Section III, we elucidate the design concepts of the proposed framework. In Section IV, the trial's specifics are explained along with the findings. Section V presents insights into performing real-world HRI tasks. In conclusion, Section VI finishes by reviewing the main facts in this study and highlighting its impact for further research.

II. ANDROID IN HRI

This study examines the android with the same task as our previous study [5]. In this section, we outline the task, robotic system, challenges, and our prior findings.

A. Task Description

For comparison purposes, we adopted the same task as in our previous study: an android robot acted as a mall receptionist to encourage visitors to use a hand sanitizer. The experiment was located in an intersection near a food court in LaLaport EXPOCITY in Osaka, and visitors could approach the android from three disparate routes.

The sanitizer task can be tough for robots. First, an open field is partially observable. Second, the intricate transitions in human dynamics complicate the cases even with a modest action space. Finally, the surroundings can exert a negative impact. In our case, there are three more sanitizers within 10 m of the android for visitors. In conclusion, we consider it an appropriate task to assess the android.

B. Robotic System

The android has three external sensors: i) A speaker, of which the working status is accessible, ii) A push switch, which is

triggered by a pedal-type hand sanitizer,² and iii) A 180° viewing angle webcam that produces full HD images. In practice, the maximum frame rate is limited to 2 fps.

The android reacts through seven pre-defined actions. Because the android made perfect predictions for **THANKS** (says “ありがとうございました (Thank you)” after sanitizer usage), **INITIALIZE** (stops all actions and the sound system), and **WAIT** (on standby), the three actions are programmed with simplistic rules. Therefore, the android is required to master the timings of the four actions:

- **HELLO (H)**: Android says “こんにちは (Hello)”.
- **SANITIZER (S)**: Android says “消毒液を使ってください (Please use the hand sanitizer)”.
- **AWAY (A)**: Android says “私に触れないでください (Please do not touch me)”.
- **TRACK (T)**: Android looks at the nearest person.

The elapsed time of the sounds is 2.5 s (**H**), 4 s (**S**), and 3.5 s (**THANKS** and **A**). The speaker's signal stays *on* within this period and turns *off* as soon as the sound ends.

C. Previous Results

In the previous work, we implemented an IRL-driven robot. To quantify its performance at this particular task, we defined *success rate* $\mathcal{R}_s = N_{thx}/N_{ct}$, where N_{thx} and N_{ct} are the number of sanitizer usage and customer traffic (CT) in a half-an-hour session. Based on $\mathcal{R}_s$, we compared five methods (*Training*, *Demo*, *Random*, *IRL*, *behavioral cloning (BC)*), of which the statistics are listed in Tables I, III, IV, V, from which we can draw the following conclusions:

- The operator's skill improves significantly throughout the training procedure (*Demo* > *Training*).
- Owing to the biased dataset, *BC* failed to learn and ended up barely better than *Random*.
- *IRL* surpassed the apprentice operator (*Training*), and behaved competitively with an expert (*Demo*).

Moreover, there was no significant difference in $\mathcal{R}_s$ between working (W) and non-working (NW) days for good policies, i.e., *Demo* and *IRL*. This is mainly because decent policies performed well regardless of CT, while bad ones frequently failed with large CT, such as in NW. This finding allows us to assess models in $\mathcal{R}_s$ regardless of W or NW under the premise that it performs on par with *Demo* and *IRL*.

III. CROSS-MODAL IRL-TO-RL TRANSITION

In this section, we first review the merits and drawbacks of *IRL*, and then describe the motivations and the specifics of the proposed learning framework.

A. Revisit IRL Policy

At time t , an android received sensor signals $l_t = [l_t^1, l_t^2]$, RGB image o_t , and human features f_t (*full-body, poses, face, head pose*). To form a state s_t , the latest four representations were stacked such that $s_t = F^{(t-3, t-2, t-1, t)}$ where $F^{(t)} = [f_t, l_t]$. Hence, a transition pair was $(s_t, a_t, r_{t+1}^{IRL}, s_{t+1})$ where $r_{t+1}^{IRL} = R^{IRL}(s_t, a_t, s_{t+1})$ and R^{IRL} was *IRL*'s reward network. Eventually, *IRL* policy $\mathcal{M}_{IRL}$ with parameter ψ behaved according to the Q value $q_t^{IRL} = Q^{IRL}(s_t, a_t; \psi)$.

²To ensure sufficient ethanol for each spray, we keep the container filled to prevent repeated presses from one individual.

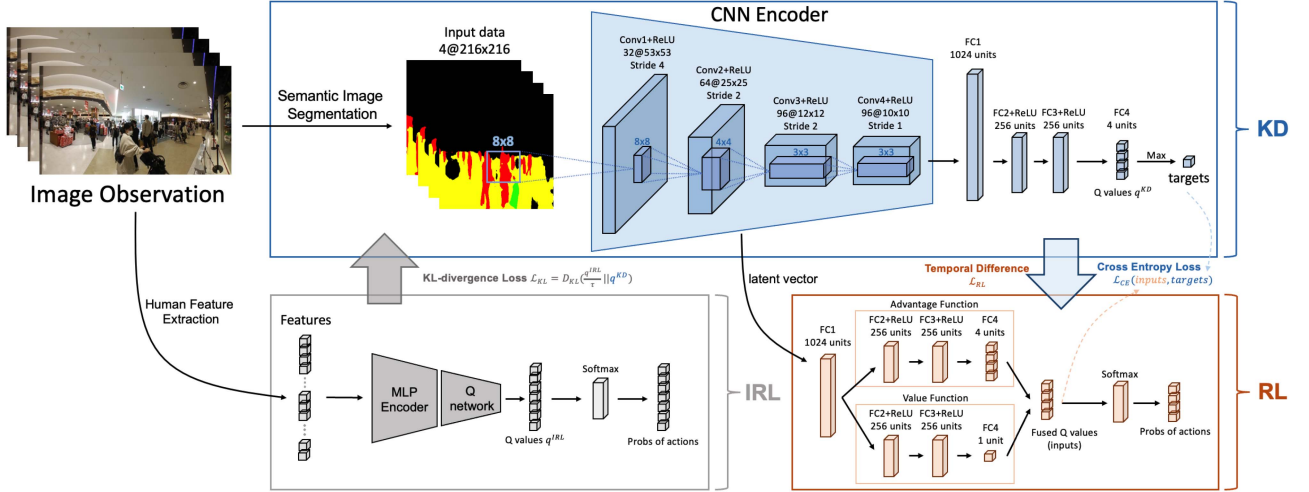


Fig. 2. Cross-modal IRL-to-RL transition architecture. IRL adopts the same setting as in [5]. KD and RL share the CNN encoder, of which the kernel size, number of channels, stride, and output resolution of each CNN layer are listed.

TABLE I
DESCRIPTIVE STATISTICS FOR THE SUCCESS RATE OF SANITIZER USAGE

Date	Initialized Weight	α	Replay Buffer	Obtained Weight	Mean	SD	Similarity ¹ train	test
—	—	—	—	$\mathcal{M}_{Training}$	.050	.020	—	—
—	—	—	—	$\mathcal{M}_{Demo}$	.070	.025	—	—
—	—	—	—	$\mathcal{M}_{Random}$	.009	.011	—	—
—	—	—	$\mathcal{D}_{demo}, \mathcal{D}_{random}$	$\mathcal{M}_{IRL}$	.059	.019	100%	100%
—	—	—	$\mathcal{D}_{demo}, \mathcal{D}_{random}$	$\mathcal{M}_{BC}$	.025	.014	—	—
—	$\mathcal{M}_{IRL}$	—	$\mathcal{D}_{demo}, \mathcal{D}_{IRL}$	$\mathcal{M}_{KD}$	—	—	92%	64%
—	$\mathcal{M}_{KD}$	$\alpha_0 = 1$	$\mathcal{D}_{demo}, \mathcal{D}_{IRL}$	$\mathcal{M}_{KD-RL}^0$	.0548	.014	88%	60%
1st	$\mathcal{M}_{KD-RL}^0$	$\alpha_1 = 0.8$	$\mathcal{D}_{demo}, \mathcal{D}_{IRL}, \mathcal{D}_1$	$\mathcal{M}_{KD-RL}^1$	.0552	.015	85%	57%
2nd	$\mathcal{M}_{KD-RL}^1$	$\alpha_2 = 0.6$	$\mathcal{D}_{demo}, \mathcal{D}_{IRL}, \mathcal{D}_1, \mathcal{D}_2$	$\mathcal{M}_{KD-RL}^2$	.069	.019	83%	57%
3rd	$\mathcal{M}_{KD-RL}^2$	$\alpha_3 = 0.4$	$\mathcal{D}_{demo}, \mathcal{D}_{IRL}, \mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3$	$\mathcal{M}_{KD-RL}^3$	.073	.025	81%	56%
4th	$\mathcal{M}_{KD-RL}^3$	$\alpha_4 = 0.2$	$\mathcal{D}_{demo}, \mathcal{D}_{IRL}, \mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3, \mathcal{D}_4$	$\mathcal{M}_{KD-RL}^4$	.088	.040	77%	56%
5th	$\mathcal{M}_{KD-RL}^4$	$\alpha_5 = 0$	$\mathcal{D}_{demo}, \mathcal{D}_{IRL}, \mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3, \mathcal{D}_4, \mathcal{D}_5$	$\mathcal{M}_{RL}$	.098	.018	58%	48%
6th	$\mathcal{M}_{RL}$	—	—	—	—	—	—	—
7th	$\mathcal{M}_{KD}$	$\alpha_7 \sim CONST$	$\mathcal{D}_{demo}, \mathcal{D}_{IRL}, \mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3, \mathcal{D}_4, \mathcal{D}_5$	$\mathcal{M}_{RL}^{CONST}$	.053	.024	43%	42%
8th	$\mathcal{M}_{KD}$	$\alpha_8 \sim Linear1$	$\mathcal{D}_{demo}, \mathcal{D}_{IRL}, \mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3, \mathcal{D}_4, \mathcal{D}_5$	$\mathcal{M}_{RL}^{Linear1}$	.076	.014	50%	46%
9th	$\mathcal{M}_{KD}$	$\alpha_9 \sim EXP$	$\mathcal{D}_{demo}, \mathcal{D}_{IRL}, \mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3, \mathcal{D}_4, \mathcal{D}_5$	$\mathcal{M}_{RL}^{EXP}$	.075	.015	45%	44%
10th	$\mathcal{M}_{KD}$	$\alpha_{10} \sim Linear2$	$\mathcal{D}_{demo}, \mathcal{D}_{IRL}, \mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3, \mathcal{D}_4, \mathcal{D}_5$	$\mathcal{M}_{RL}^{Linear2}$	.063	.015	43%	42%

¹ Similarity between IRL policy and new policies on $\mathcal{D}_{demo} + \mathcal{D}_{IRL}$ dataset, which is divided into train and test dataset for models like $\mathcal{M}_{KD}$. Previous results are included as well.

TABLE II
DECAY MODES OF α WITH $\alpha\mathcal{L}_{CE}/\mathcal{L}_{total}$ RATIO

Pattern	Decay Function	$\mathcal{L}_{total}$	$\alpha\mathcal{L}_{CE}$	Ratio
CONST	$\alpha = 0.001$	.154	.0008	.52%
Linear1	$\alpha = \begin{cases} -.0000999x + 1, & x \in [0, 1e^4] \\ 0.001, & x \in [1e^4, 2e^4] \end{cases}$	.025	.001	4.9%
EXP	$\alpha = \begin{cases} e^{-.00069087x}, & x \in [0, 1e^4] \\ 0.001, & x \in [1e^4, 2e^4] \end{cases}$	.155	.0006	.39%
Linear2	$\alpha = \begin{cases} -.00069087x + 1, & x \in [0, 1.446] \\ 0.001, & x \in [1.446, 2e^4] \end{cases}$	.125	.0007	.56%

IRL reached near-expert level despite the obstacles mentioned in Paragraph 2 of Section I. From our observation, learning from human features was a major factor that contributed to

IRL's success. Despite the modest size of the training dataset, it excluded a massive amount of irrelevant details and simplified the task for the android. However, as discussed in Section III-B, this modality has drawbacks and imposes restrictions on IRL. Therefore, switching modalities may be a viable solution to upgrade the android.

B. Switching Modality

Here, we adopted semantic image segmentation as a new visual modality, which has recently been utilized in real-world HRI settings [13]. Following the successful practices, we switched the modality for the causes below:

- Simpler to deploy: Integrating multiple detection modules is labor-intensive and unstable.

TABLE III
T-TEST RESULTS ON THE SUCCESS RATE OF SANITIZER USAGE OF DIFFERENT MODELS

Models	$\mathcal{M}_{KD-RL}^0$	$\mathcal{M}_{KD-RL}^1$	$\mathcal{M}_{KD-RL}^2$	$\mathcal{M}_{KD-RL}^3$	$\mathcal{M}_{KD-RL}^4$	$\mathcal{M}_{RL}$	$\mathcal{M}_{RL}^{CONST}$	$\mathcal{M}_{RL}^{Linear1}$	$\mathcal{M}_{RL}^{EXP}$	$\mathcal{M}_{RL}^{Linear2}$
$\mathcal{M}_{Demo}$	†	†	N.S.	N.S.	N.S.	**	†	N.S.	N.S.	N.S.
$\mathcal{M}_{KD-RL}^0$	—	N.S.	†	†	*	***	N.S.	**	**	N.S.
$\mathcal{M}_{KD-RL}^1$	—	—	N.S.	†	*	***	N.S.	**	*	N.S.
$\mathcal{M}_{KD-RL}^2$	—	—	—	N.S.	N.S.	**	N.S.	N.S.	N.S.	N.S.
$\mathcal{M}_{KD-RL}^3$	—	—	—	—	N.S.	*	N.S.	N.S.	N.S.	N.S.
$\mathcal{M}_{KD-RL}^4$	—	—	—	—	—	N.S.	*	N.S.	N.S.	†
$\mathcal{M}_{RL}$	—	—	—	—	—	—	***	*	*	***
$\mathcal{M}_{RL}^{CONST}$	—	—	—	—	—	—	—	*	*	N.S.
$\mathcal{M}_{RL}^{Linear1}$	—	—	—	—	—	—	—	—	N.S.	†
$\mathcal{M}_{RL}^{EXP}$	—	—	—	—	—	—	—	—	—	†

† $p < .1$, * $p < .05$, ** $p < .01$, *** $p < .001$, **** $p < .0001$

Results on the success rate of sanitizer usage of different models.

TABLE IV
DESCRIPTIVE STATISTICS FOR THE NORMALIZED SUM OF THE REWARD VALUES

	Training	Demo	IRL	BC	1st	2nd	3rd	4th	5th	6th	7th	8th	9th	10th
Mean	.040	.054	.039	.009	.032	.040	.050	.055	.059	.061	.028	.059	.056	.042
SD	.021	.023	.011	.008	.014	.011	.015	.025	.026	.021	.021	.013	.014	.014

Descriptive statistics for the normalized sum of the reward values.

TABLE V
STATISTICS OF HELLO AND SANITIZER

Date	Models	N_H/N_{ct}	N_S/N_{ct}	N_H/N_S	$\mathcal{R}_s$
W (NW)	$\mathcal{M}_{Demo}$	0.266 (0.219)	0.128 (0.101)	2.072 (2.170)	0.063 (0.072)
W (NW)	$\mathcal{M}_{IRL}$	0.838 (0.407)	0.129 (0.097)	6.477 (4.197)	0.057 (0.059)
W	$\mathcal{M}_{KD-RL}^0$	0.808	0.140	5.778	0.0548
NW	$\mathcal{M}_{KD-RL}^1$	0.567	0.098	5.807	0.0552
	$\mathcal{M}_{KD-RL}^2$	0.549	0.108	5.081	0.069
	$\mathcal{M}_{KD-RL}^3$	0.576	0.116	4.972	0.073
	$\mathcal{M}_{KD-RL}^4$	0.962	0.197	4.894	0.088
W	$\mathcal{M}_{RL}$	0.875	0.279	3.137	0.098
	$\mathcal{M}_{RL}^{CONST}$	0.978	0.174	5.605	0.053
	$\mathcal{M}_{RL}^{Linear1}$	0.310	0.176	1.757	0.076
W	$\mathcal{M}_{RL}^{EXP}$	0.260	0.200	1.298	0.075
	$\mathcal{M}_{RL}^{Linear2}$	0.291	0.216	1.348	0.063

Statistics of hello and sanitizer.

- Applicable to an arbitrary number of individuals without causing network inputs to explode in size.
- Extra information: In addition to humans, semantic segmentation offers knowledge of objects and backgrounds.
- Sufficient data source: In the previous study, a large amount of data was collected, which allowed the creation of an image-based learning system.

With semantic image segmentation, we anticipate the android to perceive better and thus perform better.

C. Interpreting the Proposed Framework

As *IRL*'s performance approaches that of the expert, it will encounter a bottleneck where massive amounts of new data only provide marginal advances. To sustain the rapid growth in performance, a straightforward idea is training RL on top of

IRL. Nevertheless, several issues remain unsettled: What is the design principle for a reward function? How should we take advantage of *IRL*'s prior knowledge? Is there a guarantee for continued growth in performance? Hence, we constructed the learning framework by emphasizing the following:

- * KD for switching modality.
- * Explore strategies to leverage *IRL*'s prior knowledge.
- * Confirm the feasibility of training RL with a sparse reward function using the framework.
- * Exhibit the android's capability to learn incrementally.

For the aforementioned objectives, we break the framework into two stages: 1) Transferring *IRL*'s knowledge to a cross-modal intermediary model, and 2) Using the intermediary model as a base, conducting subsequent RL with a sparse reward function and supervision from prior knowledge.

D. Cross-Modal Knowledge Distillation

Consider an intermediary model $\mathcal{M}_{KD}$ that consists of a convolutional neural network (CNN) and a fully connected (FC) network with parameter θ and ξ respectively. Distinct from $\mathcal{M}_{IRL}$, $\mathcal{M}_{KD}$ receives image segmentation from G as inputs that $g_t = G(o_t)$, where g_t is a cross-modal paired sample with f_t . As introduced in Section II-B, the simplified mechanism no longer requires the sensor inputs. The output of the intermediary model $\mathcal{M}_{KD}$ can thus be expressed as $q_t^{KD} = Q^{KD}(s'_t, a_t; \theta, \xi)$, where s'_t is stacked by g_{t-3} , g_{t-2} , g_{t-1} , and g_t . To transfer the knowledge from $\mathcal{M}_{IRL}$ to $\mathcal{M}_{KD}$, dataset $\mathcal{D}^{KD} = \{(o_i, f_i, l_i, g_i, q_i^{IRL})\}_{i=0}^{N_1}$ is prepared, in which f_i and g_i are both generated from o_i . As consistent with the findings in [14], Kullback–Leibler (KL) divergence was the best optimization objective, of which the comparison results are

elaborated in Section IV-B3:

$$\mathcal{L}_{KD}(\mathcal{D}^{KD}, \theta, \xi) = \sum_i \sigma\left(\frac{q_i^{IRL}}{\tau}\right) \ln\left(\frac{\sigma\left(\frac{q_i^{IRL}}{\tau}\right)}{\sigma(q_i^{KD})}\right) \quad (1)$$

where $\sigma(\cdot) = \text{softmax}(\cdot)$ and τ is a constant. After training, an intermediary model that inherits *IRL*'s knowledge was acquired. Furthermore, $\mathcal{M}_{KD}$ became an image-based model by switching its modality to semantic segmentation.

In the second stage, we froze the CNN's parameter θ and only trained the RL policy, as inspired by [15]. Decoupling the representation learning from the RL policy makes it more empirical to ensure a stable learning process.

E. From IRL to RL

Consider an *RL* model $\mathcal{M}_{RL}$ that inherits the CNN of $\mathcal{M}_{KD}$. $\mathcal{M}_{RL}$ initializes a policy network with parameter ϕ_{RL} under the Rainbow architecture³ [16]. The android acted based on the Q value $q_i^{RL} = Q^{RL}(s'_i, a_i; \theta, \phi_{RL})$. To guide the android, a sparse reward function R^{RL} was hand-crafted which generated feedback $r_{t+1}^{RL} = R^{RL}(s_t, a_t, s_{t+1})$. Hence, the *RL* policy learned through Q-learning by optimizing the 1-step and n-step temporal difference (TD) that

$$\mathcal{L}_{RL}(\mathcal{D}^{RL}, \phi_{RL}|\theta) = \mathcal{L}_1 + \lambda \mathcal{L}_n, \quad (2)$$

$$\mathcal{L}_1 = \mathbb{E}\left([r_{t+1}^{RL} + \gamma Q'(s_{t+1}, a'_{t+1}) - Q^{RL}(s_t, a_t)]^2\right), \quad (3)$$

$$\mathcal{L}_n = \mathbb{E}\left([r_{t+1:t+n}^{RL} + \gamma^n Q'(s_{t+n}, a'_{t+n}) - Q^{RL}(s_t, a_t)]^2\right), \quad (4)$$

where constant $\lambda, \gamma \in [0, 1]$, Q' is the target network of Q^{RL} , $a'_{t+j} = \arg\max_a Q^{RL}(s_{t+j}, a)$, $r_{t+1:t+n}^{RL} = \sum_{k=t+1}^{t+n} \gamma^{k-t-1} r_k$. Moreover, $\mathcal{D}^{RL} = \{(g_i, a_i, r_{i+1}^{RL}, g_{i+1})\}_{i=0}^{N_2}$ contains data in $\mathcal{D}^{KD}$ and newly collected data.

To leverage the prior knowledge, $\mathcal{M}_{RL}$ was supervised by $\mathcal{M}_{KD}$'s hard labels a_i^{KD} using cross-entropy (CE) loss:

$$\mathcal{L}_{CE}(\mathcal{D}^{CE}, \phi_{RL}|\theta, \xi) = - \sum_i a_i^{KD} \log q_i^{RL} \quad (5)$$

where $\mathcal{D}^{CE} = \{(g_i, a_i^{KD})\}_{i=0}^{N_3}$. Because the scale of $\mathcal{M}_{KD}$'s and $\mathcal{M}_{RL}$'s Q functions are unrelated, it becomes viable to leverage $\mathcal{M}_{KD}$'s policy with (5). Consequently, $\mathcal{M}_{RL}$ optimized the below objective in the second stage:

$$\mathcal{L}_{total} = \mathcal{L}_{RL} + \alpha \mathcal{L}_{CE} \quad (6)$$

where α can be a constant or a decaying variable.

F. Related Work

With the inherent constraints of imitation learning (IL), it is arduous for apprentices to reach the expert level, especially with limited demonstration and high interaction costs. Hence, integrating IL with RL is a viable approach for enhancing robots' performance. In [17], a typical IRL+RL paradigm was explored offline where IRL first inferred a reward function in preparation for RL's next use. Such a manner leverages the merits of both IRL and RL. However, IRL's expertise is not fully utilized, as RL may also benefit from IRL's policy network. In [18], a BC+RL objective was investigated where RL was trained online under BC's supervision. Nevertheless, a huge amount of interaction steps are required. Here, our proposed framework: 1) obtained the knowledge of modality with the aid of IRL, 2) supervised RL with an intermediary policy, 3) warm-started RL with the previously gathered data, and 4) awarded RL with a sparse reward function.

IV. REFINING THE BEHAVIOR POLICY

This section elaborates on the training schedule, model architecture, implementation, and results. This section also introduces an additional evaluation indicator.

A. Training Schedule

Here, we performed a ten-day trial. Each day, the android was tested in nine sessions for 4.5 h. The experiment was divided into three parts. First, pre-training was conducted to leverage the existing data. Second, we spent the first six days verifying the proposed framework. Finally, we investigated α 's decay mode in the last four days and suggested a preferable mode based on our empirical experience.

1) *Prior to the Trial*: $\mathcal{M}_{KD-RL}^0$ was obtained before the experiment. To make the most of the available dataset, $\mathcal{M}_{KD}$ was first obtained by cross-modal KD. Then, $\mathcal{M}_{KD-RL}^0$ was acquired by subsequent RL with $\alpha_0 = 1$. In this stage, the training dataset $\mathcal{D} = [\mathcal{D}_{demo}, \mathcal{D}_{IRL}]$ where $\mathcal{D}_{demo}$ and $\mathcal{D}_{IRL}$ were data collected in the previous study.

2) *The First 6 Days*: Each day, the android behaved with the initialized weight, gathered new data, and then was trained with a decaying α for the obtained weight. As listed in Table I, the android started with $\mathcal{M}_{KD-RL}^0$ on Day 1, and finished with $\mathcal{M}_{RL}$ on Day 5. Meanwhile, α decayed day by day that $\alpha_i = 1 - 0.2i$ on the i -th day. Moreover, the training data became $\mathcal{D} = [\mathcal{D}_{demo}, \mathcal{D}_{IRL}, \mathcal{D}_1, \dots, \mathcal{D}_i]$, that $\mathcal{D}_i$ was collected on Day i . On Day 6, the android finished transitioning from *IRL* to *RL*, and *RL* was evaluated.

3) *The Last 4 Days*: Contrast to the arithmetic progression, four more decay modes were tested. The decay functions for *CONST*, *Linear1*, *EXP*, and *Linear2* are defined in Table II, in which x is the training step. The training data is $\mathcal{D} = [\mathcal{D}_{demo}, \mathcal{D}_{IRL}, \mathcal{D}_1, \dots, \mathcal{D}_5]$. As exponential distribution (EXP) cannot decay to 0, α 's minimum was set to 0.001, assuming that the CE loss is ignorable in this value. According to the table, the ratio of weighted CE loss over total loss remained at an extremely low level, which supports our assumption.

B. Experiment Details

1) *Semantic Image Segmentation*: SegFormer [19] was employed and trained on ADE20 K dataset [20], which features

³ Among all the extensions, distributional RL and noisy nets were unused.

annotations for daily necessities. In practice, objects were classified into four categories⁴: *Person*, *Objects*, *Floor*, and *Background*, to which grayscale values (0.299, 0.587, 0.114, and 0) were assigned. For image pre-processing, a Full HD image was first resized to $216 \times 384 \times 3$ and recognized by SegFormer. Then, it was converted to a gray image and rescaled to 216×216 . At last, we stacked the latest four observations as the representation. On average, the process consumed 0.2 s on a GeForce RTX 2080 Super GPU.

2) *Data Preparation*: The data in the previous study were carried over: $\mathcal{D}_{demo}$ (85609, **H**:2%, **S**:1%) and $\mathcal{D}_{IRL}$ (130302, **H**:6%, **S**:1%). The values in parentheses, in that order, are the number of all actions, and the proportion of **H** and **S**. In addition, 119689 new samples were collected from Day 1 to Day 5. During the training process, data were randomly separated into 90% train and 10% test datasets.

3) *Knowledge Distillation*: $\mathcal{M}_{KD}$ performs the best when $\tau=0.1$ (accuracy: train-91.58%, test-64.15%). In addition to (1), mean-square-error (MSE) loss and intrinsic reward approach (IR) [21] were examined. However, both MSE (accuracy: train-70.04%, test-57.30%) and IR (accuracy: train-36.49%, test-33.14%) methods fell short of KL-divergence loss in terms of performance.

4) *Data Relabeling*: As stated in Section IV-B2, **H** and **S** merely account for a small portion because action selection is prohibited until the chosen one is finished. To gain more access to IRL's expertise, $\mathcal{D}^{KD}$ was relabeled using $\mathcal{M}_{IRL}$ with the sensor signals zeroed ($l_i = \mathbf{O}$). Thus, the relabeled $\mathcal{D}_{demo}$ (**H**:50%, **S**:14%) and $\mathcal{D}_{IRL}$ (**H**:41%, **S**:12%) offered $\mathcal{M}_{KD}$ more insights into $\mathcal{M}_{IRL}$. Furthermore, **TRACK**'s Q value was replaced with the selected actions if the chosen one is **THANKS**, **INITIALIZE**, or **WAIT**.

5) *Reward Function*: For simplicity, we consider a reward function that merely provides feedback to **SANITIZER**:

$$r_{i+1}^{RL} = \begin{cases} 0 & \text{if } a_i \neq \mathbf{S} \\ -0.1 & \text{if } a_i = \mathbf{S} \text{ and } N=0 \\ N & \text{if } a_i = \mathbf{S} \text{ and } N>0 \end{cases} \quad (7)$$

where N is the usage's count within up to T timesteps after a **S** is called and before the next **S** is triggered. Hence, a valid **S** ($N>0$) is rewarded whereas an invalid one ($N=0$) is punished. In practice, we set $T=40$ for: i) Usage from multiperson; ii) Extended period of contemplation and hesitation; iii) Need for assistance from others during usage.

6) *Architecture and Hyper-Parameters*: The architectures of the models are depicted in Fig. 2. For IRL, we adopted the same settings as in the previous study. Moreover, $\mathcal{M}_{RL}$ inherited the architecture of $\mathcal{M}_{IRL}$'s policy network. For the hyper-parameters, the batch size is 1280, the learning rate is 3×10^{-4} , $\gamma=0.9$, and $\lambda=1$. For the learning steps, $\mathcal{M}_{KD}$ early-stopped at 73490 steps; $\mathcal{M}_{KD-RL}^0$, $\mathcal{M}_{KD-RL}^1$, $\mathcal{M}_{KD-RL}^2$, $\mathcal{M}_{KD-RL}^3$, $\mathcal{M}_{KD-RL}^4$ finished learning at 40000 steps; As the dataset expanded, $\mathcal{M}_{RL}$, $\mathcal{M}_{RL}^{CONST}$, $\mathcal{M}_{RL}^{Linear1}$, $\mathcal{M}_{RL}^{EXP}$, $\mathcal{M}_{RL}^{Linear2}$ were only trained for 20000 steps.

C. Evaluation of Android's Behavior

1) *Customer Traffic*: The experiment was done for ten consecutive days (4W+6NW). To obtain CT, we employed ByteTrack [22], which identified visitors whose sizes exceeded a

⁴*Objects*: 23 classes (carpet, cushion, pillow, flower, book, computer, fan, arcade machine, towel, pole, bottle, toy, barrel, basket, bag, cradle, ball, food, blanket, tray, plate, drinking glass, flag). *Background*: 125 classes.

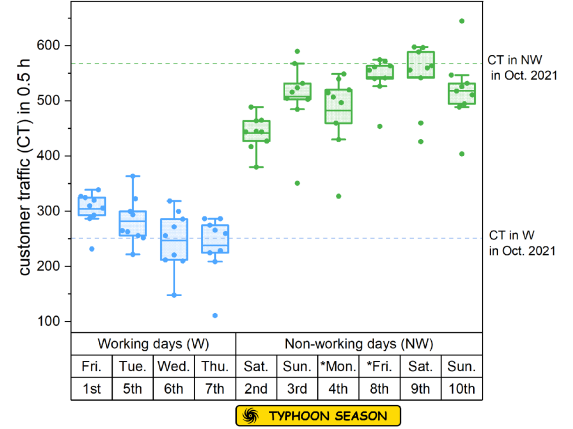


Fig. 3. Customer traffic throughout the experiment.

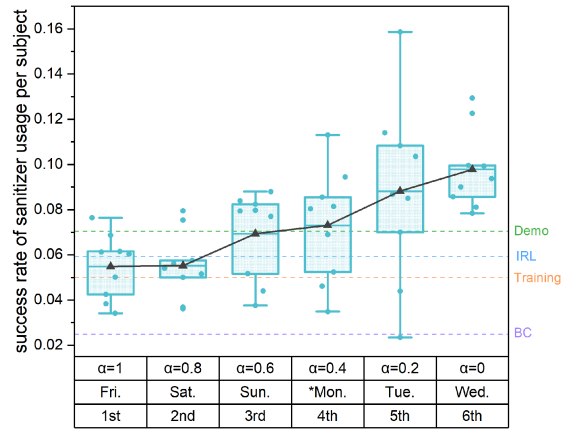


Fig. 4. Success rate of sanitizer usage in the first six days.

certain threshold (i.e., 10000). Because visitors who rapidly passed by were also counted, N_{ct} is larger than the number of participants who engaged in the interaction. Fig. 3 depicts CT's boxplots in days, which are classified into W or NW. Meanwhile, all days of the week are noted and holidays other than weekends are marked with asterisks. In addition, two lines are drawn to calibrate the average CT in October 2021. Because a typhoon warning was in effect, we also marked the worst days of the typhoon. Fortunately, no significant difference was found in CT between October 2021 and September 2022 in both W ($p<.1$) and NW ($p<.01$) using two-sample two-tailed t-tests.

2) *Success Rate*: Figs. 4 and 5 display the boxplots for $\mathcal{R}_s$. Furthermore, α 's values and the matching decay modes are noted in the figures. To facilitate comparison, we also exhibit the mean values of earlier models as baselines for the new models. More statistics for the baselines and the new models can be found in Tables I and III.

In the first six days, the android learned under the framework with α decreasing according to arithmetic progression. Throughout the trial, the android performed better and better day by day. On Day 1, we discovered that $\mathcal{M}_{KD-RL}^0$'s performance was superior to $\mathcal{M}_{Training}$'s but inferior to $\mathcal{M}_{IRL}$'s. The cause can be a loss of knowledge during the distillation. As the android proceeded to learn, it began to surpass $\mathcal{M}_{IRL}$ and reached the human-expert level ($p=0.91$) on Day 3. Eventually, it outperformed $\mathcal{M}_{Demo}$ on Day 6, the day when the policy completed the transition to RL. Meanwhile, the statistics revealed the models'

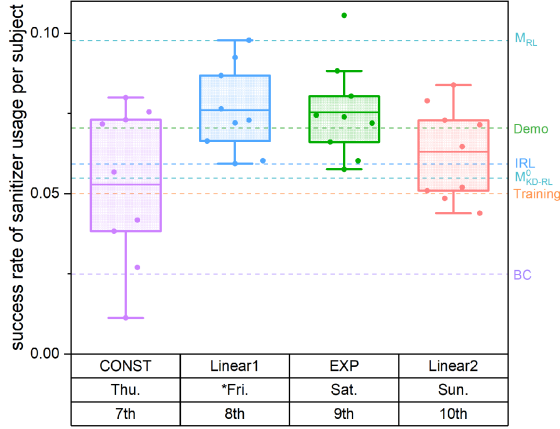


Fig. 5. Success rate of sanitizer usage in the last four days.

progress. Based on the t-test results, new policies did not differ significantly from the adjacent days, but remarkable improvement could be observed two days apart. Moreover, the similarity to *IRL* slowly declined as learning progressed. This meets our expectations, because a drop in similarity may be interpreted as a catastrophic loss of prior knowledge and lead to tremendous uncertainties. Thus, the findings demonstrate that the proposed framework enabled the android to perform a seamless and robust transition from *IRL* to *RL*.

In the last four days, four decay modes of α were tested; three of them reached human-expert level except $\mathcal{M}_{RL}^{CONST}$. Based on the average $\mathcal{R}_s$, the policies were prioritized in this order: $\mathcal{M}_{RL}^{Linear1}$, $\mathcal{M}_{RL}^{EXP}$, $\mathcal{M}_{RL}^{Linear2}$, and $\mathcal{M}_{RL}^{CONST}$. Among these, pairs (*Linear1*, *EXP*) and (*CONST*, *Linear2*) have comparable performances. To explore a proper decay mode, we compared *EXP*+*Linear2* and found that moderate decline in the late stage is essential with the premise of rapid decline in the early stage. Moreover, the contrast between *EXP/Linear1* revealed that the decline speed in the early stage had little impact on performance. Compared to prior models, *CONST* was worse than *IRL* and $\mathcal{M}_{KD-RL}^0$, indicating that using *RL* with a sparse reward function and without the supervision of prior knowledge could destroy the policy. Furthermore, *RL* outperformed each of the four decay modes. One of the potential causes is insufficient training due to time constraints that a huge disparity exists in training steps (*RL*: 220 *k* vs. $\mathcal{M}_{RL}^*$: 20 *k*). As affected by α 's rapid decline, $\mathcal{M}_{RL}^*$'s similarity to *IRL* is significantly lower than *RL*.

D. Additional Evaluation Indicator

We defined the sum of rewards $\mathcal{S}_r = \sum_j r_j^{RL}$, which adds up the rewards in a half-an-hour session. Unlike $\mathcal{R}_s$ measuring the overall performance, $\mathcal{S}_r$ evaluates by calculating the cumulative reward that i) merely offers positive feedback to a valid **S**, ii) punishes an invalid **S**, and iii) ignores usages that might be irrelevant to the android. Because $\mathcal{S}_r$ was impacted by CT, the normalized sum of rewards $\hat{\mathcal{S}}_r = \mathcal{S}_r / N_{ct}$ was introduced, which is upper-bounded by $\mathcal{R}_s$.

Fig. 6 and Table IV exhibit the statistics of $\hat{\mathcal{S}}_r$. First, minor changes in previous models' results occur in comparison to $\mathcal{R}_s$: i) *IRL* was downgraded such that a significant difference with *Demo* was found ($p < .01$), ii) *Training* turned better such that it had no significant difference from *Demo* ($p = 0.14$). Second, the android kept upgrading in the first six days, but with slight

adjustments. Although the android reached *Demo*-level on Day 3 ($p = 0.61$), the superiority over *Demo* remained invisible on Day 6 ($p = 0.43$). Moreover, the android exceeded *IRL* on Day 2 and *Training* on Day 3. Finally, $\hat{\mathcal{S}}_r$'s results are consistent with $\mathcal{R}_s$ in the last four days. Because excessive triggers of **S** will be punished, the results suggest that the android activates actions at appropriate timings. In addition, the consistency between $\hat{\mathcal{S}}_r$ and $\mathcal{R}_s$ indicates that the android has a substantial effect on visitors' behaviors.

V. DISCUSSION

A. Android's Presence and Task Performance

Although the android attracted visitors' attention and induced them to approach, its presence does not necessarily promote task performance for the following reasons:

- No explicit disclosure of the android's intention: While a hand sanitizer was placed, one may take it for granted solely for virus prevention.
- Excessive expectations of functionality: Visitors frequently approached the android in hopes of communicating but left disappointed. Owing to the mismatch of expectations, visitors became reluctant to cooperate.

B. Welcome and Ask

The interplay between **H** and **S** played a critical role in an interaction. Table V presents the statistics of the two actions, in which N_H and N_S are numbers of executions. According to the previous results, the models tend to trigger less **H** in NW while little variation exists in N_S/N_{ct} between W and NW. In the first six days, N_H/N_S remained in decline while $\mathcal{R}_s$ rose. This can be explained by the daily increase in N_S and the timing-appropriate activation of extra **S**. In addition, *RL* triggers 3.3 times more **H**, 2.2 times more **S**, and is 1.6 times more superior than *Demo* in $\mathcal{R}_s$ in W. However, while more "Welcome and Ask" may be beneficial, mastering the timings is hard, even for a human operator who is reluctant to bother visitors with excessive actions. Our results indicate that the android developed a more proactive and effective strategy under the proposed framework, whereas *CONST* fails with such an aggressive approach. Moreover, we discovered that maintaining N_H/N_S between 2 and 3 is preferable for interacting with visitors in the task.

C. Irl-to-Better

1) *Modality*: Perception is onerous in real-world HRI. For broader applications, the ideal case is acquiring as much information with affordable devices. Hence, webcams are usually more practical than Lidars or high-end audio devices. Because images are typically cluttered with irrelevant details, feature extraction becomes essential. However, weighing the pro and cons of various visual modalities is arduous. For instance, human-feature recognition is detailed but redundant, whereas semantic segmentation is concise but sketchy. Fusing multiple modalities may be considered a promising solution for robots to leverage the visual input.

2) *Decay of the Supervision*: Owing to the expensive cost of experimenting in the real world, only a few tests were conducted to investigate α 's decay mode. Based on the findings, a preferable mode drops in the early stage and moderates in the late

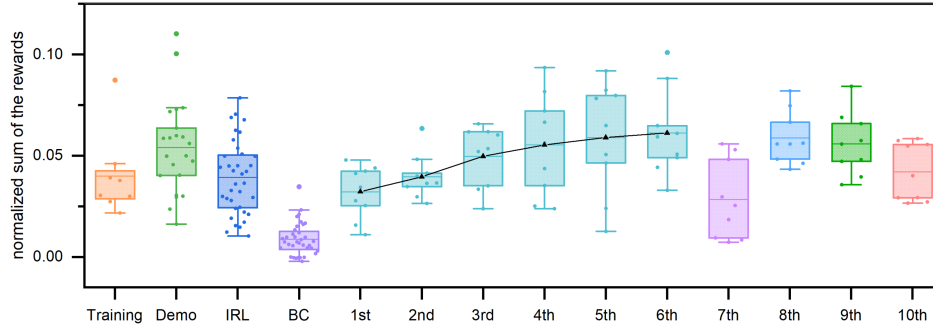


Fig. 6. Normalized sum of the reward values, of which the mean values from 1st to 6th are marked with triangles.

stage. By adopting the suggested decay mode, we anticipate the training process of robots being shortened while the performance maintains rapid growth.

3) *Knowledge Loss*: Throughout the knowledge transfer, it is inevitable that loss will occur, although fresh data may lessen the impact. Hence, exploring approaches to prevent such loss is essential. Here, the android begins with the latest weight and is guided by $\mathcal{M}_{KD}$ on each training day. Such a configuration can be adjusted to leverage the knowledge of accessible assets. For instance, the android's decisions may be formed from the previously acquired models' weighted sum. Moreover, the android may be supervised by some newly obtained models during training. These strategies may alleviate knowledge loss and enhance robots' performance.

4) *RL or Otherwise*: This study attempts to enhance the android's skills in a real-world HRI task. As a single method has limitations, integrating various approaches with distinct attributes into a learning framework offers great potential to boost robots' performance. Therefore, the proposed framework applies to target models other than *RL*. Furthermore, the framework is expected to benefit robots from a broader perspective, enabling more X-to-Y transition applications.

VI. CONCLUSION

This study explored a feasible technique to upgrade an IRL-driven android in real-world HRI. Although IRL can rapidly master skills by LfD, it will encounter a bottleneck as the agent approaches that of the expert. To overcome the bottleneck, we propose a cross-modal IRL-to-RL transition framework. Based on a near-expert IRL policy, the framework allows subsequent RL using a sparse reward function. To demonstrate its validity, we conducted a ten-day field experiment. Our findings indicate that the framework enables the android to maintain rapid growth in performance and eventually surpass the human teacher. In conclusion, this letter offers a viable approach for refining well-behaved robots in real-world HRI. We anticipate it can considerably assist robots in mastering human skills. Moreover, we wish our framework to inspire more IRL-to-better learning paradigms, facilitating robots to exceed the expert level in more real-world HRI applications.

REFERENCES

- [1] C. Finn, S. Levine, and P. Abbeel, "Guided cost learning: Deep inverse optimal control via policy optimization," in *Proc. Int. Conf. Mach. Learn.*, 2016, pp. 49–58.
- [2] J. Ho and S. Ermon, "Generative adversarial imitation learning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2016, vol. 29, pp. 4565–4573.
- [3] D. Hadfield-Menell, S. J. Russell, P. Abbeel, and A. Dragan, "Cooperative inverse reinforcement learning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2016, pp. 3909–3917.
- [4] H. Kretschmar, M. Spies, C. Sprunk, and W. Burgard, "Socially compliant mobile robot navigation via inverse reinforcement learning," *Int. J. Robot. Res.*, vol. 35, no. 11, pp. 1289–1307, 2016.
- [5] Z. Chen, Y. Nakamura, and H. Ishiguro, "Android as a receptionist in a shopping mall using inverse reinforcement learning," *IEEE Robot. Automat. Lett.*, vol. 7, no. 3, pp. 7091–7098, Jul. 2022.
- [6] D. S. Brown, W. Goo, and S. Niekum, "Better-than-demonstrator imitation learning via automatically-ranked demonstrations," in *Proc. Conf. Robot Learn.*, 2020, pp. 330–359.
- [7] D. Silver et al., "A general reinforcement learning algorithm that masters chess, shogi, and Go through self-play," *Science*, vol. 362, no. 6419, pp. 1140–1144, 2018.
- [8] P. R. Wurman et al., "Outracing champion Gran Turismo drivers with deep reinforcement learning," *Nature*, vol. 602, no. 7896, pp. 223–228, 2022.
- [9] V. Mnih et al., "Human-level control through deep reinforcement learning," *Nature*, vol. 518, no. 7540, pp. 529–533, 2015.
- [10] I. Akkaya et al., "Solving rubik's cube with a robot hand," 2019, *arXiv:1910.07113*.
- [11] C. Chen, Y. Liu, S. Kreiss, and A. Alahi, "Crowd-robot interaction: Crowd-aware robot navigation with attention-based deep reinforcement learning," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2019, pp. 6015–6022.
- [12] H. Modares, I. Ranatunga, F. L. Lewis, and D. O. Popa, "Optimized assistive human-robot interaction using reinforcement learning," *IEEE Trans. Cybern.*, vol. 46, no. 3, pp. 655–667, Mar. 2016.
- [13] M. Siam et al., "Video object segmentation using teacher-student adaptation in a human robot interaction (HRI) setting," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2019, pp. 50–56.
- [14] A. A. Rusu et al., "Policy distillation," in *Proc. Int. Conf. Learn. Representations*, 2016, pp. 1–13.
- [15] R. M. Shah and V. Kumar, "RRL: Resnet as representation for reinforcement learning," in *Proc. Int. Conf. Mach. Learn.*, M. Meila and T. Zhang, Eds., 2021, vol. 139, pp. 9465–9476.
- [16] M. Hessel et al., "Rainbow: Combining improvements in deep reinforcement learning," in *Proc. AAAI Conf. Artif. Intell.*, 2018, pp. 3215–3222.
- [17] I. Halperin, J. Liu, and X. Zhang, "Combining reinforcement learning and inverse reinforcement learning for asset allocation recommendations," 2022, *arXiv:2201.01874*.
- [18] Y. Lu et al., "Imitation is not enough: Robustifying imitation with reinforcement learning for challenging driving scenarios," 2022, *arXiv:2212.11419*.
- [19] E. Xie, W. Wang, Z. Yu, A. Anandkumar, J. M. Alvarez, and P. Luo, "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, vol. 34, pp. 12077–12090.
- [20] B. Zhou, H. Zhao, X. Puig, S. Fidler, A. Barriuso, and A. Torralba, "Scene parsing through ADE20K dataset," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 5122–5130.
- [21] W. M. Czarnecki, R. Pascanu, S. Osindero, S. M. Jayakumar, G. Swirszcz, and M. Jaderberg, "Distilling policy distillation," in *Proc. Int. Conf. Artif. Intell. Statist.*, 2019, pp. 1331–1340.
- [22] Y. Zhang et al., "Bytetrack: Multi-object tracking by associating every detection box," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 1–21.